FDI and Growth: Panel Evidence with Income-Group Heterogeneity

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2026-03-31

Abstract

This paper examines the relationship between foreign direct investment (FDI) and host-country economic growth using an unbalanced panel of 193 economies over the period 1991 to 2024. To mitigate reverse causality, FDI is lagged by one period and two-way fixed effects are employed that absorb time-invariant country characteristics and global time trends. A conditional convergence control following Barro (1991) further reduces omitted-variable bias. The main contribution is an income-group interaction that tests Findlay's (1978) relative-backwardness hypothesis: whether technology spillovers from FDI are larger where the gap with the global frontier is greatest. The results suggest that the aggregate FDI-growth relationship is attenuated once fixed effects are introduced, but this attenuation masks income-group heterogeneity. In above-median-income economies the FDI coefficient is statistically indistinguishable from zero. In below-median-income economies a one-percentage-point increase in the lagged FDI-to-GDP ratio is associated with approximately 0.27 percentage points of additional annual GDP per capita growth, significant at the 0.1% level. These findings are consistent with technology-transfer mechanisms and suggest that the income-group heterogeneity in FDI-growth effects has practical relevance for host-country industrial and investment policy.

1 Introduction

In recent decades, governments worldwide have competed to attract foreign direct investment (FDI), offering tax incentives, special economic zones, and investment protection treaties in the belief that inward capital generates jobs, transfers technology, and accelerates economic growth. Yet the empirical evidence on whether FDI delivers these gains remains contested. While the literature documents positive correlations between FDI and various development outcomes, establishing causality is harder: faster-growing, more stable economies also attract more FDI, potentially reversing the direction of causation.

The distinction between these two interpretations has policy consequences. The case of Costa Rica is cited as an illustration of FDI-led structural transformation: from the

*Replication materials: github.com/dufourlorenzo/replication-fdi-growth-panel

1990s onward, the country attracted FDI inflows in manufacturing and services, and this experience is often held up as evidence for the potential of export-oriented FDI (Yamuni et al., 2023). Whether FDI drove that transformation or reflected Costa Rica's prior investments in human capital and institutional stability is disputed. If FDI causes growth, investment promotion agencies represent an efficient use of public resources; if FDI is primarily reactive, those resources may be better directed at improving fundamentals.

This paper contributes evidence by exploiting within-country temporal variation in an unbalanced panel covering 193 economies from 1991 to 2024. Three methodological choices distinguish the approach. First, FDI is lagged by one period relative to growth, which mitigates the most direct form of simultaneity bias. Second, two-way fixed effects are employed, absorbing time-invariant country characteristics (geography, colonial history, institutional quality) and year-specific global shocks, so that identification draws on within-country changes in FDI over time. Third, the analysis tests for income-group heterogeneity in the FDI-growth relationship. The theoretical prior, dating to Findlay (1978), is that technology spillovers from FDI are largest where the technology gap between the host economy and the global frontier is greatest, precisely in lower-income countries. Testing this mechanism requires both a long panel and the income-group interaction employed here.

The main results are as follows. The pooled cross-sectional correlation between FDI and growth is positive and large, confirming earlier findings in the literature. Once country and year fixed effects are introduced, however, the average effect falls, consistent with level selection (wealthier countries attracting more FDI for reasons unrelated to FDI's own effect on growth). For above-median-income economies, the fixed-effects estimate is statistically indistinguishable from zero. For below-median-income economies the total effect is approximately 0.59 percentage points of GDP per capita growth per unit of asinh(FDI) ($p < 0.001$). These findings are consistent with Findlay (1978)'s relative-backwardness hypothesis and with the pattern documented by Pasali & Chaudhary (2020), who find that the effects of foreign ownership on firm performance vary substantially by firm size and host-country context.

The remainder of the paper is organised as follows. Section 2 reviews the relevant theoretical and empirical literature. Section 3 describes the data. Section 4 lays out the empirical strategy. Section 5 presents and discusses the results. Section 6 concludes with policy implications and directions for future research.

2 Background

2.1 Theoretical Foundations

The economics of FDI and growth has been shaped by three main theoretical traditions. The first concerns the motivations for multinational activity. Dunning (1979) synthesises earlier contributions in the "OLI" (Ownership-Location-Internalisation) eclectic framework: a firm invests abroad when it holds firm-specific ownership advantages (technology,

brand, management know-how), when exploiting them in a foreign location offers productivity or market benefits, and when internal organisation of cross-border transactions is cheaper than licensing. The OLI paradigm emphasises that FDI is primarily a vehicle for firms to exploit, not develop, their advantages; yet the host country may benefit from knowledge spillovers as local firms and workers interact with foreign affiliates.

Two theoretical frameworks motivate the income-heterogeneity prediction tested in this paper. Findlay (1978) derives that the rate of technological catching-up in the host country is increasing in the technology gap between the foreign and domestic sectors: countries furthest from the global frontier should experience the largest productivity gains from each unit of inward FDI, a result known as the "relative-backwardness hypothesis." Wang & Blomström (1992) model technology diffusion as a function of the multinational's investment in transferring its technology stock and the host economy's absorptive capacity. Their framework implies that spillovers are attenuated when absorptive capacity is too low, suggesting gains accrue primarily to economies above a minimum absorptive-capacity threshold. These two mechanisms generate a tension: Findlay predicts larger gains for lower-income (higher technology-gap) countries, while Wang and Blomström caution that the lowest-income economies may lack the absorptive capacity to capture those gains. The income-group interaction estimated below provides a reduced-form test of whether below-median-income economies in the aggregate benefit more from FDI than above-median-income economies; it cannot cleanly distinguish between the two mechanisms and should be read as a test of the net effect of income on FDI-growth sensitivity.

The growth literature provides the second building block. Barro (1991)'s cross-country evidence on conditional convergence establishes that, conditional on structural characteristics, poorer countries grow faster, consistent with neoclassical diminishing returns to capital. This motivates including initial income as a control variable: without it, the negative correlation between income levels and growth would confound any variable that is itself correlated with income. In the two-way fixed-effects specification, the log of lagged GDP per capita proxies for a country's distance from its own steady state, and the recovered coefficient ($\hat{\delta} \approx -3.78$) provides a useful specification check. Note that in a within-country fixed-effects framework this coefficient captures within-country mean reversion rather than the cross-country convergence documented by Barro (1991); the negative sign is expected under the Solow mechanism but should not be overinterpreted as confirming it.

2.2 Empirical Literature

The empirical literature on FDI and growth is large and yields mixed results. Kechagia & Metaxas (2018) survey sixty years of research and document a wide range of findings: many studies identify positive effects, but a significant share find negligible or negative relationships, with magnitudes varying considerably across samples and identification strategies. Borensztein et al. (1998) find that FDI raises growth only in countries with sufficient human capital, providing early evidence for an absorptive-capacity threshold.

Alfaro et al. (2004) show that local financial market development is a further conditioning factor. Carkovic & Levine (2005) apply GMM estimators to a large cross-country panel and find little evidence of an independent causal effect of FDI on growth once country fixed effects are controlled for, highlighting the identification challenge.

Narula & Driffield (2012) argue that the empirical evidence on FDI-assisted growth is more ambiguous than widely acknowledged, pointing to reverse causality, heterogeneous spillover conditions, and the inadequacy of aggregate FDI data as persistent sources of bias. The approach here addresses reverse causality through lagged regressors and fixed effects, though a valid instrument would be needed to fully resolve the endogeneity problem.

Blomström (1991) catalogues the channels—competitive pressure, labour training, and technology transfer to local suppliers—through which host-country firms may benefit from multinational presence, and notes that spillover gains are larger where local firms have the skill base to absorb foreign technologies. Tomohara & Takii (2011) document positive wage effects of FDI in manufacturing, consistent with labour-market spillovers.

2.3 Contribution

This paper makes two contributions. Methodologically, it uses a two-way fixed-effects estimator with country-clustered standard errors on a panel of 193 economies, which is more demanding than the pooled or cross-section designs common in the earlier literature. Substantively, it tests the Findlay (1978) heterogeneity prediction using an income-group interaction in a within-country framework, rather than relying on cross-sectional income variation that is confounded with other country characteristics. The main finding is that the FDI-growth relationship is statistically indistinguishable from zero for above-median-income economies and positive for below-median-income economies, consistent with the relative-backwardness hypothesis, though the identification strategy falls short of the IV benchmark required for a causal interpretation.

3 Data

3.1 Sources and Variables

The analysis draws on three annual series from the World Bank Development Indicators (World Bank, 2026):

1. FDI net inflows as a percentage of GDP (series `BX.KLT.DINV.WD.GD.ZS`): the explanatory variable.
2. GDP per capita growth, annual % (series `NY.GDP.PCAP.KD.ZG`): the dependent variable.
3. GDP per capita, PPP (constant 2021 international $) (series `NY.GDP.PCAP.PP.KD`): used to construct the convergence control and the income-group indicator.

A fourth series, trade openness (exports plus imports as a share of GDP, series `NE.TRD.GNFS.ZS`), is used in robustness check R7.

The PPP-adjusted constant-price series is preferred over nominal current-USD GDP per capita for two reasons. First, PPP adjustment removes cross-country price-level differences, ensuring that the income-group classification reflects real purchasing power rather than nominal exchange rates. Second, using constant prices removes inflation effects, so that the convergence control and income-group indicator reflect real income changes over time.

After removing regional and income-group aggregates from the World Bank's reporting structure, the raw data cover 193 individual economies. The panel is unbalanced because FDI reporting begins at different dates for different countries; the earliest observations are from 1991 and the latest from 2024.

3.2 Data Cleaning

All estimation uses R (R Core Team, 2025) with packages WDI (Arel-Bundock et al., 2019), fixest (Bergé, 2018), tidyverse (Wickham et al., 2019), and kableExtra (Zhu, 2024).

Three decisions require explicit justification. First, both the dependent variable and the FDI regressor are winsorised at the 1st and 99th percentiles of their respective pooled distributions. Raw growth rates include episodes of conflict and statistical revisions generating values below -50% or above 100%, and FDI-to-GDP can exceed 1,000% in small holding-company hubs—reflecting balance-sheet intermediation rather than real investment. Winsorising growth to [-14.5%, 16.8%] and FDI to [-6.6%, 52%] retains all country-year observations while bounding these extremes.

Second, the FDI series contains 1.3% zeros and 8.7% negative values. Zeros arise in years when no investment inflows were recorded (common in countries with thin financial sectors), while negative values reflect disinvestment and profit repatriation. The natural logarithm cannot handle these values: $\log(x)$ is undefined for $x \leq 0$. Dropping zeros and negatives would remove precisely the low-FDI, often lower-income observations most relevant to the heterogeneity analysis, inducing selection bias. The inverse hyperbolic sine (IHS) transformation $\text{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$ avoids this problem: it is defined for all real x, equals zero when $x = 0$, handles negative values symmetrically, and approximates $\ln(2x)$ for large positive x (Bellemare & Wichman, 2020). Coefficients on asinh(FDI) are interpretable as semi-elasticities, similarly to log-transformed regressors.

Third, the income-group indicator D_i^L classifies each country as below- or above-median by comparing its time-averaged GDP per capita to the cross-country median of those averages. This produces 96 below-median economies out of 193 total, with a median threshold of approximately \$13k in constant 2021 international dollars (PPP). The time-averaging removes cyclical variation and classifies countries by their structural income level rather than their position in any given year.

Table 1: Summary Statistics: Estimation Sample

Table 2: Summary statistics for the three core variables in the estimation sample. Both growth and FDI are winsorised at the 1st and 99th percentiles before model estimation. FDI is reported after the IHS transformation is applied to the winsorised lagged raw value.

Variable	Mean	SD	Min	Max	N
GDPpc Growth (%)	2.04	4.65	-14.49	16.77	6109
asinh(FDI), lagged	1.54	1.25	-2.58	4.64	6109
Log GDPpc, lagged	9.35	1.19	6.24	12.07	6109

3.3 Summary Statistics

Table 1 shows that the average GDP per capita growth rate in the sample is 2%, with a standard deviation of 4.7 percentage points after winsorisation. The IHS-transformed FDI variable has a mean of 1.5 and a maximum of 4.6. The mean log GDP per capita of 9.35 corresponds to approximately $11.5k in constant 2021 international dollars (PPP).

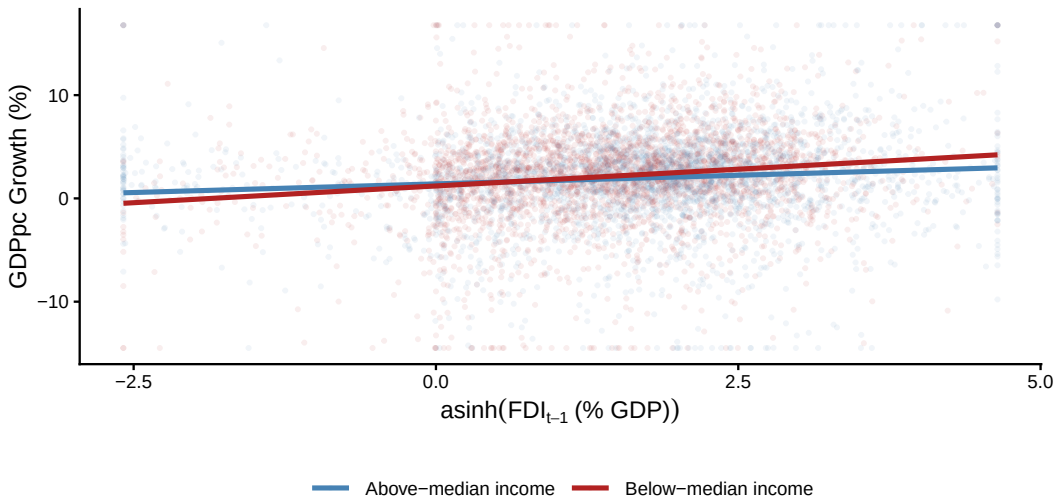

Figure 1: IHS-transformed lagged FDI (x-axis) and GDP per capita growth (y-axis). Both variables are winsorised at the 1st and 99th percentiles. Lines are separate OLS fits by income group.

Figure 1 plots the raw relationship. A steeper gradient is visible for below-median-income countries, consistent with the heterogeneity that motivates the main specification.

4 Methodology

4.1 Identification Challenges

Two main threats complicate causal inference. The first is reverse causality: multinational investors target growing, politically stable economies, so that a positive FDI-growth correlation may reflect growth causing FDI rather than FDI causing growth. One-period

lagging of the regressor mitigates this concern without eliminating it entirely.

The second challenge is omitted variables. Countries that attract more FDI may also implement institutional reforms, liberalise trade, or invest in infrastructure in ways that independently raise growth. Unobservable time-invariant characteristics (geography, legal origins, colonial history) may simultaneously attract FDI and correlate with long-run growth levels. Country fixed effects absorb all such time-invariant confounders. Year fixed effects remove any global shock (commodity price cycles, global financial crises, pandemic effects) that affects all countries simultaneously and is correlated with both global FDI flows and global growth.

A third, less easily addressed threat is time-varying omitted variables. Country fixed effects absorb any determinant of growth that is constant within a country, but not variables that change over time: trade liberalisation episodes, domestic financial deepening, institutional reform, or infrastructure investment may all be correlated with FDI inflows and independently affect growth. The specification includes only one time-varying control (lagged income) beyond FDI, so time-varying confounders remain a residual concern. This is an acknowledged limitation of the design, and results should be interpreted as conditional associations rather than causal estimates.

4.2 Model Specifications

Specification 1: Pooled OLS Baseline. The most transparent starting point is a pooled regression with no controls beyond the lagged FDI variable:

$$g_{it} = \alpha + \beta_1 \operatorname{asinh}(\mathrm{FDI}_{i,t-1}) + \varepsilon_{it} \tag{1}$$

where g_{it} denotes annual GDP per capita growth for country i in year t. This benchmark captures the unconditional correlation and is comparable to the bulk of the early literature.

Specification 2: Two-Way Fixed Effects. To exploit within-country temporal variation and control for omitted country-level factors, country (α_i) and year (λ_t) fixed effects are added:

$$g_{it} = \alpha_i + \lambda_t + \beta_1 \operatorname{asinh}(\mathrm{FDI}_{i,t-1}) + \varepsilon_{it} \tag{2}$$

The coefficient $\hat{\beta}_1$ is identified from within-country deviations: the estimator removes each country's time mean and each year's cross-country mean, so variation in FDI that is common to all years for a given country (level differences) and variation common to all countries in a given year (global cycles) are both absorbed. What remains is the idiosyncratic change in a country's FDI position relative to its own history, net of the common year effect. This sweeps out confounding from any time-invariant determinant of growth.

Specification 3: Conditional Convergence Control. Following Barro (1991), the log of lagged GDP per capita is added as a control for conditional convergence:

$$g_{it} = \alpha_i + \lambda_t + \beta_1 \operatorname{asinh}(\mathrm{FDI}_{i,t-1}) + \delta \ln \mathrm{GDPpc}_{i,t-1} + \varepsilon_{it} \tag{3}$$

A negative $\hat{\delta}$ would confirm that, conditional on fixed effects, countries further from their steady state grow faster, a standard check that the specification captures the convergence mechanism documented in the growth literature.

Specification 4: Income-Group Heterogeneity. The primary specification augments Equation 3 with an interaction term:

$$g_{it} = \alpha_i + \lambda_t + \beta_1 \operatorname{asinh}(\mathrm{FDI}_{i,t-1}) + \beta_2 \big[\operatorname{asinh}(\mathrm{FDI}_{i,t-1}) \times D_i^L\big] + \delta \ln \mathrm{GDPpc}_{i,t-1} + \varepsilon_{it} \tag{4}$$

where $D_i^L = 1$ if country i's time-averaged GDP per capita falls below the cross-country sample median. The country fixed effects absorb the (time-invariant) main effect of D_i^L. The interaction coefficient $\hat{\beta}_2$ captures the additional FDI-growth sensitivity of below-median-income economies relative to above-median-income economies within the same panel. Under the relative-backwardness hypothesis (Findlay, 1978), $\hat{\beta}_2 > 0$, though it should be noted that the within-country identification cannot fully isolate the Findlay mechanism from other income-related heterogeneity.

Standard errors are clustered by country throughout, accounting for arbitrary serial correlation within the country time series. Section 5.3 reports seven robustness checks that directly address the limitations of this baseline: (i) a time-varying income classification that re-classifies countries each year against the annual cross-country median (addressing the convergence-country misclassification concern); (ii) a balanced panel restricting to countries with at least 20 annual observations; (iii) alternative lag structures at $t-2$ and $t-3$; (iv) two-way country-and-year clustered standard errors; (v) an initial-period income classification that avoids look-ahead bias; (vi) exclusion of financial centres and resource enclaves with distorted FDI-to-GDP ratios; and (vii) a trade openness control.

5 Results

5.1 Regression Estimates

Table 2 presents the four specifications, discussed in turn below.

Column (1): Pooled OLS. The unconditional association is positive: $\hat{\beta}_1 = 0.452$ ($p < 0.001$), implying that a one-unit increase in asinh(FDI) is associated with 0.45 percentage points of additional annual growth. This is consistent with the positive correlations reported in the survey literature (Kechagia & Metaxas, 2018). This estimate likely overstates any causal effect due to omitted country heterogeneity.

Table 3: FDI and GDP Per Capita Growth: Panel Estimates

Dependent Variable:	GDPpc Growth			
	Pooled OLS	Two-way FE		
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	1.34***			
	(0.150)			
asinh FDI(t-1)	0.452***	0.337***	0.316***	0.134
	(0.077)	(0.089)	(0.079)	(0.098)
ln GDPpc(t-1)			-3.78***	-3.81***
			(0.461)	(0.454)
asinh FDI(t-1) × Below-median				0.457***
				(0.157)
Fixed-effects				
code		Yes	Yes	Yes
year		Yes	Yes	Yes
Fit statistics				
Observations	6,109	6,109	6,109	6,109
R^2	0.01481	0.26275	0.28812	0.29052

Clustered (code) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Country-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Below-median dummy equals 1 if country's time-averaged GDP per capita (PPP, constant 2021 international $) is below the cross-country sample median. Country and year fixed effects absorbed in columns (2) to (4).

Column (2): Two-Way FE without convergence control. Adding country and year fixed effects reduces the estimate to $\hat{\beta}_1 = 0.337$, still statistically significant. The 25% reduction from column (1) is consistent with a level-selection effect: countries with higher FDI tend to be wealthier and to grow faster for reasons unrelated to FDI.

Column (3): Two-Way FE with convergence control. The addition of $\ln \mathrm{GDPpc}_{i,t-1}$ leaves the FDI coefficient unchanged at 0.316. The convergence coefficient $\hat{\delta} \approx -3.78$ ($p < 0.001$) is negative: a 10% higher initial income is associated with approximately 0.36 pp less annual growth, conditional on fixed effects, consistent with Barro (1991).

Column (4): Income-group heterogeneity. The results differ across income groups. For above-median-income economies, the FDI semi-elasticity falls to $\hat{\beta}_1 = 0.134$, which is statistically indistinguishable from zero. For below-median-income economies the total effect is $\hat{\beta}_1 + \hat{\beta}_2 \approx 0.591$ ($p < 0.001$). The positive $\hat{\beta}_2$ is consistent with the pattern predicted by Findlay (1978).

5.2 Consistency with Theory

The income-group heterogeneity in column (4) is consistent with the Findlay (1978) argument at the macroeconomic level. A one-percentage-point increase in the lagged FDI-to-GDP ratio is associated with approximately 0.27 pp more annual GDP per capita growth in below-median-income economies, evaluated at their median FDI share of 2%.[1] Both the technology-gap mechanism (Findlay (1978)) and the capital-scarcity mechanism (Dunning (1979)) would predict larger gains in lower-income countries with shallow domestic financial systems, while neither would reliably produce the positive effect at the frontier. The statistically insignificant result for above-median-income economies is also consistent with competitive displacement by multinationals, though the design cannot separate these explanations; that remains a question for future work.

5.3 Robustness Checks

Table 3 summarises seven robustness exercises (R1–R7), the third of which comprises two lag variants (t-2 and t-3). The baseline result is qualitatively unchanged across all specifications.

R1: Time-varying income classification. The static D_i^L dummy assigns countries to a group for their entire history, potentially misclassifying rapidly converging economies (e.g., South Korea, Singapore) in later decades. To address this, D_i^L is replaced with a time-varying indicator $D_{it}^L = \mathbf{1}[\mathrm{GDPpc}_{it} < \mathrm{median}_j(\mathrm{GDPpc}_{jt})]$ that re-classifies each country against the annual cross-country median. The total below-median-income effect under this specification is 0.344 ($p < 0.01$), smaller in magnitude than the baseline (0.591) but still positive and statistically significant, suggesting the result is not an artefact of time-fixed group assignment. The attenuation is consistent with the time-varying classification

[1] Formally, $\partial g_{it}/\partial \mathrm{FDI}_{i,t-1} \approx (\hat{\beta}_1 + \hat{\beta}_2)/\sqrt{1 + \mathrm{FDI}_{\mathrm{med}}^2} \approx 0.265$.

Table 4: Robustness Checks: Income-Group FDI Effects across Alternative Specifications

Table 5: Robustness of the income-heterogeneity result. Stars on both columns: *p<0.10, **p<0.05, ***p<0.01 (country-clustered SE). Above-median $\hat{\beta}$ is the coefficient on asinh(FDI_{t-k}); Below-median total effect is $\hat{\beta}_1 + \hat{\beta}_2$ tested via Wald test. R4 uses same point estimates as the baseline but reclusters SE by country and year simultaneously. N for R3a and R3b are smaller because the additional lag removes the first one or two years per country. R5 classifies countries by GDP per capita in their first sample year. R6 excludes countries with mean FDI/GDP > 10%. R7 adds lagged trade openness and is estimated on the subsample where trade data are available.

Specification	N	**Above-median $\hat{\beta}$**	**Below-median total effect**
Baseline (m4): static income groups, country-clustered SE	6,109	0.134	0.591***
R1: Time-varying income classification (annual median)	6,109	0.296***	0.344***
R2: Balanced panel ($\geq$ 20 obs/country; $n_{\text{cty}} = 182$)	5,985	0.157	0.586***
R3a: Two-year lag, asinh(FDI_{t-2})	5,915	0.130	0.528***
R3b: Three-year lag, asinh(FDI_{t-3})	5,723	0.083	0.376***
R4: Two-way clustered SE (country × year)	6,109	0.134	0.591***
R5: Initial-period income classification ($n_{\text{low}} = 96$)	6,109	0.090	0.614***
R6: Exclude financial centres and resource enclaves ($n_{\text{excl}} = 20$)	5,509	0.137	0.533***
R7: Trade openness control (lagged trade/GDP)	5,231	0.179**	0.574***

more aggressively reassigning converging countries to the above-median group in later periods.

R2: Balanced panel. The main sample is an unbalanced panel: countries with short data histories contribute only a handful of observations and could disproportionately drive the results. To assess this, the sample is restricted to the 182 countries with at least 20 annual observations, ensuring that each country's fixed effect is estimated from a meaningful number of time periods. The total below-median-income effect under this restriction is 0.586 ($p < 0.001$), consistent with the baseline result not being driven by the unbalanced structure.

R3: Alternative lag structures. To assess whether the one-period lag is sufficient to address timing concerns, the full specification is re-estimated using two-year ($t-2$) and three-year ($t-3$) lags of FDI. Each additional lag removes the first year of each country's time series, reducing the sample to 5,915 and 5,723 observations for $t-2$ and $t-3$, respectively. The total below-median-income FDI effects are 0.528 ($p < 0.001$) and 0.376 ($p < 0.01$), respectively. The gradual attenuation from $t-1$ to $t-3$ is consistent with FDI spillovers materialising within the first couple of years following investment.

R4: Two-way clustered standard errors. Country-only clustering may under-estimate standard errors if unobserved shocks produce cross-sectional correlation within years (e.g., global capital flow surges). Re-clustering on both country and year simultaneously raises the standard error on the above-median-income FDI coefficient to 0.104 and on the interaction to 0.17, but the total below-median-income effect remains significant at the $0.1% level.

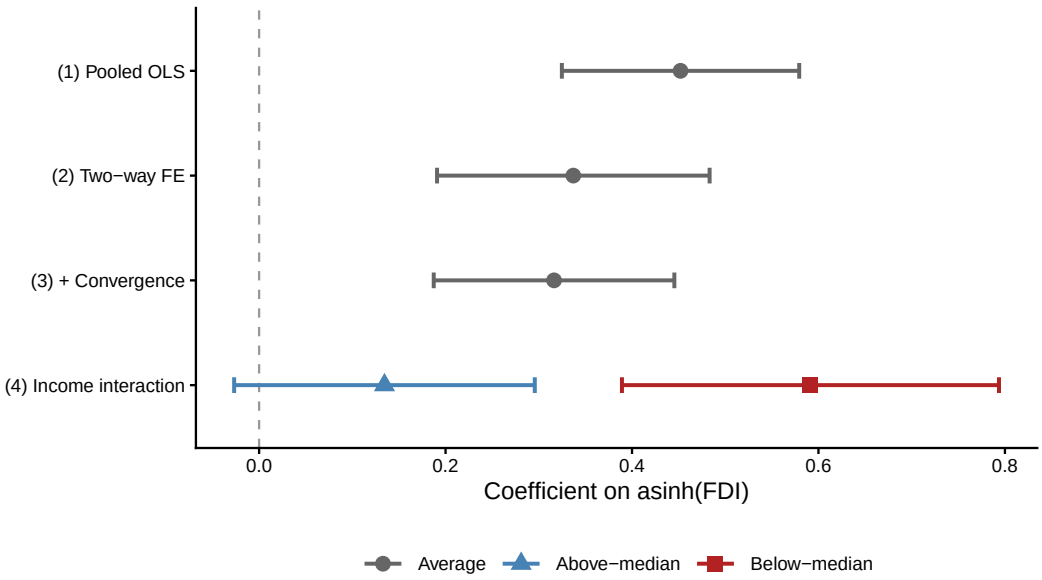

Figure 2: FDI coefficient estimates with 90% confidence intervals across specifications. For the income-interaction model (Spec. 4) two estimates are shown: the above-median effect ($\hat{\beta}_1$) and the total below-median effect ($\hat{\beta}_1 + \hat{\beta}_2$). Confidence intervals based on country-clustered standard errors.

Figure 2 plots the coefficient estimates across all four main specifications with 90% confidence intervals. For below-median-income economies the estimate is positive and statistically different from zero across all specifications that separate income groups.

R5: Initial-period income classification. The static D_i^L dummy based on time-averaged income introduces look-ahead bias: countries that grew slowly during the sample are more likely to be classified as below-median, and that slow growth is also the outcome being explained. To address this, each country is classified using only its GDP per capita in the first year it appears in the estimation sample, relative to the cross-country median of those initial values. This classification is fully prospective. Under this criterion 96 economies are classified as initially below-median. The total below-median-income FDI effect is 0.614 ($p < 0.001$), confirming that the heterogeneity result does not depend on the backward-looking income classification.

R6: Exclude financial centres and resource enclaves. FDI-to-GDP ratios exceeding 10% on average typically reflect balance-sheet intermediation in offshore financial centres or large single-project resource extraction, rather than productive investment. The 20 countries with mean FDI/GDP above this threshold are excluded (Cayman Islands, Malta, Hong Kong SAR, China, Luxembourg, Cyprus, and 15 others). The remaining 5,509 observations yield an above-median estimate of 0.137 (n.s.) and a below-median total effect of 0.533 ($p < 0.001$), unchanged from the baseline.

R7: Trade openness control. Trade-open economies may simultaneously attract more FDI and grow faster, so that the FDI coefficient could partly proxy for trade rather than capturing FDI's own effect. Adding lagged trade (exports plus imports as a share of

GDP) as a time-varying control—estimated on the 5,231 observations for which this series is available—leaves the above-median estimate at 0.179 ($p < 0.05$) and the below-median total effect at 0.574 ($p < 0.001$), suggesting trade openness is not the primary driver of the result.

Despite these checks, two structural limitations remain. First, the residual endogeneity concern cannot be fully resolved without a valid instrument. Bilateral investment treaty ratification networks from the EDIT database (Alschner et al., 2021) represent a natural candidate, covering 4,143 agreements across 258 economies. However, every instrument variant constructed from the EDIT data yields first-stage F statistics well below the conventional threshold of 10 for the main FDI equation. The failure appears structural: annual within-country variation in BIT stock is insufficient to move aggregate FDI-to-GDP, and the remaining variation is absorbed by year fixed effects that capture the global BIT proliferation wave of the 1990s and 2000s. A valid IV strategy requires bilateral FDI data matched to treaty ratification, where the treaty-to-flow link can be measured directly. Second, the aggregate FDI measure conflates resource-seeking, market-seeking, and efficiency-seeking investment (Dunning, 1979), as well as reinvested earnings alongside new greenfield entry; disaggregated bilateral data would allow a more detailed investigation of mechanisms.

6 Conclusion

This paper examines the relationship between foreign direct investment and GDP per capita growth using an unbalanced panel of 193 economies over 1991 to 2024. By combining one-period lagging, two-way fixed effects, and a conditional convergence control, the analysis addresses identification challenges arising from reverse causality and time-invariant omitted variables. Residual endogeneity cannot be ruled out without a valid instrument, so results are best interpreted as conditional associations rather than causal estimates. The central finding is that the aggregate FDI-growth coefficient, while positive, masks income-group heterogeneity. In above-median-income economies, within-country changes in FDI have no statistically significant effect on growth. In below-median-income economies, a one-percentage-point increase in the lagged FDI-to-GDP ratio is associated with approximately 0.27 pp of additional annual GDP per capita growth, an effect that is statistically significant.

These results are consistent with the Findlay (1978) relative-backwardness hypothesis: technology spillovers from FDI appear largest where the gap with the global frontier is greatest. The finding is robust to seven alternative specifications: replacing the static income-group classification with a time-varying annual-median criterion, restricting estimation to the balanced sub-panel of countries with at least 20 annual observations, shifting the lag structure to $t-2$ or $t-3$, tightening inference with two-way country-and-year clustered standard errors, using an initial-period income classification that avoids look-ahead bias (R5), excluding financial centres and resource enclaves with distorted

FDI-to-GDP ratios (R6), and adding a lagged trade openness control (R7). In all cases the total below-median-income FDI effect remains positive and statistically significant.

These results have implications for FDI promotion policy. For above-median-income countries, the estimated growth returns to FDI attraction are negligible on average, suggesting that fiscal incentives aimed at attracting multinationals may offer limited value as a growth strategy. For below-median-income countries, the evidence is consistent with FDI having a positive association with growth, which motivates continued attention to investment-climate conditions. These results cannot, however, identify which specific policies most effectively attract productive FDI or whether the estimated association would persist under policy-induced increases in FDI flows.

Future work should pursue a credible instrumental-variable strategy to sharpen the causal interpretation—bilateral investment treaty ratification data (Alschner et al., 2021) are a natural candidate when matched with bilateral FDI flows—decompose the aggregate FDI effect by investment type following the Dunning (1979) OLI framework, and examine whether the intensive margin (increased investment by incumbents) and the extensive margin (new entrant multinationals) carry different growth implications.

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